

Artwork: Test Image10

Scale-Invariant Coordinate System

$$\tilde{X} = 2 \left(\frac{x}{W} \right) - 1, \quad \tilde{Y} = 1 - 2 \left(\frac{y}{H} \right) \quad X = \frac{\left(\tilde{X} + \frac{2}{139} \right)}{\frac{162}{167}}, \quad Y = \frac{\left(\tilde{Y} - \frac{32}{179} \right)}{1}, \quad \theta = \arctan \left(\frac{Y}{X} \right)$$

Layer Definitions Boundary Manifolds

$$C_0(X, Y, v) = \frac{42 + v + v^2}{40} \cdot \left(1 + \frac{6}{121}X + \frac{1}{69}Y \right)$$

$$X_1 = \left(X - \frac{4}{177} \right), \quad Y_1 = \left(Y - \frac{20}{41} \right)$$

$$\tau_1(X, Y) = \frac{69}{79} \cos^2 \left(\frac{2713}{193}X_1 + \frac{749}{118}Y_1 - \frac{259}{151} \cos \left(\frac{489}{155}X_1 + \frac{86}{29}Y_1 \right) \right) \quad + \frac{103}{129} \cos^4 \left(\frac{1031}{179}X_1 + \frac{2468}{121}Y_1 + \frac{14}{17} \cos \left(\frac{606}{175}X_1 + \frac{109}{26}Y_1 \right) \right) + \frac{95}{139} \cos^6 \left(\frac{4554}{155}X_1 + \frac{352}{31}Y_1 - \frac{21}{19} \cos \left(\frac{1031}{179}X_1 + \frac{2468}{121}Y_1 + \frac{14}{17} \cos \left(\frac{606}{175}X_1 + \frac{109}{26}Y_1 \right) \right) \right)$$

$$X_2 = \left(X + \frac{7}{97} \right), \quad Y_2 = \left(Y - \frac{32}{131} \right)$$

$$\tau_2(X, Y) = \frac{1}{56} \cos^2 \left(\frac{1411}{86}X_2 + \frac{500}{79}Y_2 - \frac{15}{188} \cos \left(\frac{183}{46}X_2 + \frac{339}{187}Y_2 \right) \right) \quad + \frac{2}{35} \cos^4 \left(\frac{1237}{197}X_2 + \frac{3313}{165}Y_2 + \frac{8}{67} \cos \left(\frac{200}{99}X_2 + \frac{163}{33}Y_2 \right) \right) - \frac{31}{157} \cos^6 \left(\frac{4520}{163}X_2 + \frac{2313}{196}Y_2 + \frac{19}{124} \cos \left(\frac{1237}{197}X_2 + \frac{3313}{165}Y_2 + \frac{8}{67} \cos \left(\frac{200}{99}X_2 + \frac{163}{33}Y_2 \right) \right) \right)$$

$$X_3 = \left(X + \frac{1}{28} \right), \quad Y_3 = \left(Y - \frac{31}{104} \right)$$

$$\tau_3(X, Y) = -\frac{6}{85} \cos^2 \left(\frac{2856}{179}X_3 + \frac{960}{181}Y_3 + \frac{17}{66} \cos \left(\frac{138}{35}X_3 + \frac{64}{23}Y_3 \right) \right) \quad - \frac{1}{14} \cos^4 \left(\frac{91}{15}X_3 + \frac{1907}{96}Y_3 + \frac{40}{87} \cos \left(\frac{160}{79}X_3 + \frac{254}{59}Y_3 \right) \right) - \frac{9}{196} \cos^6 \left(\frac{5625}{197}X_3 + \frac{787}{62}Y_3 - \frac{104}{177} \cos \left(\frac{91}{15}X_3 + \frac{1907}{96}Y_3 + \frac{40}{87} \cos \left(\frac{160}{79}X_3 + \frac{254}{59}Y_3 \right) \right) \right)$$

$$X_4 = \left(X + \frac{14}{109} \right), \quad Y_4 = \left(Y + \frac{32}{157} \right)$$

$$C_4(v) = \frac{16 + v^2}{40} \Phi_4(X, Y) \quad = 1 - \left(\frac{X_4}{\frac{27}{115}} \right)^2 - \left(\frac{Y_4}{\frac{55}{147}} \right)^2 \quad - \frac{5}{86} \cos(1\theta - 0.003) - \frac{3}{38} \cos \left(2\theta + \frac{75}{148} \right) - \frac{14}{165} \cos \left(3\theta + \frac{118}{117} \right) - \frac{6}{79} \cos \left(4\theta + \frac{128}{85} \right) \quad + \frac{3}{166} \cos(5\theta + 2)$$

$$X_5 = \left(X + \frac{5}{169} \right), \quad Y_5 = \left(Y + \frac{51}{167} \right)$$

$$C_5(v) = \frac{17}{40} \Phi_5(X, Y) \quad = 1 - \left(\frac{X_5}{\frac{33}{127}} \right)^2 - \left(\frac{Y_5}{\frac{66}{169}} \right)^2 \quad + \frac{7}{159} \cos(1\theta - 0) + \frac{4}{163} \cos \left(2\theta + \frac{1}{2} \right) + \frac{3}{188} \cos(3\theta + 1) + \frac{1}{75} \cos \left(4\theta + \frac{3}{2} \right) \quad + \frac{2}{181} \cos(5\theta + 2)$$

$$X_6 = \left(X + \frac{34}{189}\right), \quad Y_6 = \left(Y + \frac{91}{172}\right)$$

$$\tau_6(X, Y) = -\frac{3}{47} \cos^2 \left(\frac{193}{12} X_6 + \frac{832}{139} Y_6 \right) - \frac{12}{187} \cos^4 \left(\frac{1113}{184} X_6 + \frac{3416}{171} Y_6 + \frac{4}{45} \cos(2X_6 + 5Y_6) \right) + \frac{13}{110} \cos^6 \left(\frac{1542}{55} X_6 + \frac{1355}{113} Y_6 \right) - \frac{2}{33} \cos^8 \left(\frac{2143}{179} X_6 + \frac{7201}{200} Y_6 \right)$$

$$X_7 = \left(X + \frac{3}{20}\right), \quad Y_7 = \left(Y + \frac{89}{193}\right)$$

$$\tau_7(X, Y) = \frac{7}{40} \cos^2 \left(\frac{3058}{193} X_7 + \frac{49}{8} Y_7 - \frac{5}{44} \cos \left(\frac{789}{194} X_7 + \frac{362}{185} Y_7 \right) \right) + \frac{16}{87} \cos^4 \left(\frac{997}{173} X_7 + \frac{3138}{155} Y_7 - \frac{49}{183} \cos \left(\frac{150}{73} X_7 + \frac{708}{143} Y_7 \right) \right) + \frac{19}{118} \cos^6 \left(\frac{4160}{149} X_7 + \frac{1095}{91} Y_7 + \frac{17}{165} \cos(2X_7 + 5Y_7) \right)$$

$$X_8 = \left(X + \frac{19}{53}\right), \quad Y_8 = \left(Y + \frac{19}{45}\right)$$

$$\tau_8(X, Y) = \frac{1}{44} \cos^2 \left(\frac{2965}{186} X_8 + \frac{535}{89} Y_8 - \frac{4}{11} \cos \left(\frac{686}{171} X_8 + \frac{41}{21} Y_8 \right) \right) + \frac{5}{171} \cos^4 \left(\frac{502}{81} X_8 + 20.001 Y_8 - \frac{100}{199} \cos \left(\frac{337}{186} X_8 + \frac{460}{99} Y_8 \right) \right) + \frac{1}{53} \cos^6 \left(\frac{3814}{137} X_8 + \frac{1486}{129} Y_8 - \frac{14}{47} \cos(2X_8 + 5Y_8) \right)$$

$$X_9 = \left(X + \frac{29}{188}\right), \quad Y_9 = \left(Y + \frac{59}{194}\right)$$

$$\tau_9(X, Y) = -\frac{12}{73} \cos^2 \left(\frac{545}{34} X_9 + 5.998 Y_9 \right) - \frac{18}{109} \cos^4 \left(\frac{1140}{191} X_9 + \frac{1861}{93} Y_9 - \frac{11}{169} \cos(2X_9 + 5Y_9) \right) - \frac{33}{199} \cos^6(28.004 X_9 + 12 Y_9) - \frac{8}{47} \cos^8 \left(\frac{868}{73} X_9 + \frac{6411}{178} Y_9 - \frac{20}{15} \cos(2X_9 + 5Y_9) \right)$$

$$X_{10} = \left(X + \frac{28}{141}\right), \quad Y_{10} = \left(Y + \frac{68}{115}\right)$$

$$\tau_{10}(X, Y) = \frac{9}{151} \cos^2 \left(\frac{1253}{76} X_{10} + \frac{954}{155} Y_{10} \right) + \frac{12}{193} \cos^4 \left(\frac{893}{154} X_{10} + \frac{1652}{83} Y_{10} - \frac{5}{17} \cos \left(\frac{317}{189} X_{10} + \frac{77}{16} Y_{10} \right) \right) + \frac{5}{88} \cos^6 \left(\frac{3621}{130} X_{10} + \frac{2239}{190} Y_{10} \right) + \frac{4}{83} \cos^8 \left(\frac{1366}{111} X_{10} + \frac{1095}{91} Y_{10} + \frac{17}{165} \cos(2X_{10} + 5Y_{10}) \right)$$

$$X_{11} = \left(X + \frac{11}{157}\right), \quad Y_{11} = \left(Y + \frac{26}{103}\right)$$

$$\tau_{11}(X, Y) = \frac{2}{83} \cos^2(16X_{11} + 6Y_{11}) + \frac{2}{83} \cos^4(6X_{11} + 20Y_{11}) + \frac{1}{123} \cos^6(28X_{11} + 12Y_{11}) + \frac{4}{167} \cos^{12}(48X_{11} + 20Y_{11}) C_{11}(X, Y, v) = \frac{22 + v + v^2}{40} \cdot \left(1 + \frac{v}{2}\right)$$

$$X_{12} = \left(X + \frac{2}{117}\right), \quad Y_{12} = \left(Y + \frac{59}{87}\right)$$

$$\tau_{12}(X, Y) = \frac{3}{188} \cos^2 \left(\frac{844}{57} X_{12} + \frac{980}{179} Y_{12} - \frac{113}{130} \cos \left(\frac{297}{74} X_{12} + \frac{164}{189} Y_{12} \right) \right) + \frac{2}{47} \cos^4 \left(\frac{202}{33} X_{12} + \frac{2619}{125} Y_{12} + \frac{59}{150} \cos \left(\frac{312}{185} X_{12} + \frac{599}{104} Y_{12} \right) \right) - \frac{1}{21} \cos^6 \left(\frac{4689}{167} X_{12} + \frac{1835}{156} Y_{12} - \frac{14}{47} \cos(2X_{12} + 5Y_{12}) \right)$$

$$X_{13} = \left(X + \frac{35}{114}\right), \quad Y_{13} = \left(Y + \frac{41}{54}\right)$$

$$\tau_{13}(X, Y) = -\frac{2}{49} \cos^2 \left(\frac{1147}{72} X_{13} + \frac{76}{13} Y_{13} + \frac{40}{193} \cos \left(\frac{699}{181} X_{13} + \frac{319}{149} Y_{13} \right) \right) - \frac{38}{185} \cos^4 \left(\frac{1096}{185} X_{13} + \frac{3517}{172} Y_{13} + \frac{84}{143} \cos \left(\frac{83}{30} X_{13} + \frac{409}{77} Y_{13} \right) \right) - \frac{14}{53} \cos^6 \left(\frac{1617}{58} X_{13} + \frac{378}{31} Y_{13} \right)$$

$$X_{14} = \left(X - \frac{1}{200}\right), \quad Y_{14} = (Y + 0.002)$$

$$\tau_{14}(X, Y) = -\frac{23}{89} \cos^2 \left(\frac{2568}{163} X_{14} + \frac{897}{185} Y_{14} + \frac{81}{182} \cos \left(\frac{411}{118} X_{14} + \frac{497}{181} Y_{14} \right) \right) - \frac{3}{14} \cos^4 \left(\frac{194}{31} X_{14} + \frac{3799}{187} Y_{14} \right) - \frac{89}{197} \cos^6 \left(\frac{1819}{65} X_{14} + \frac{2363}{198} Y_{14} - \frac{137}{190} \cos \left(\frac{475}{66} X_{14} + \frac{578}{141} Y_{14} \right) \right)$$

$$X_{15} = \left(X - \frac{1}{31}\right), \quad Y_{15} = \left(Y - \frac{2}{75}\right)$$

$$\tau_{15}(X, Y) = \frac{8}{157} \cos^2 \left(\frac{2405}{154} X_{15} + \frac{617}{106} Y_{15} + \frac{33}{86} \cos \left(\frac{300}{73} X_{15} + \frac{91}{40} Y_{15} \right) \right) + \frac{7}{184} \cos^4 \left(\frac{665}{128} X_{15} + \frac{2439}{124} Y_{15} + \frac{29}{182} \cos \left(\frac{281}{129} X_{15} + \frac{889}{172} Y_{15} \right) \right) + \frac{2}{77} \cos^6 \left(\frac{3529}{127} X_{15} + \frac{404}{33} Y_{15} \right)$$

Master Composition (Recursive Layer Synthesis)

$$H_v^{(0)}(X, Y) = C_0(X, Y, v)$$

$$H_v^1(X, Y) = (1 - W_1(X, Y)) \cdot H_v^0(X, Y) + W_1(X, Y) \cdot C_1(X, Y, v)$$

$$H_v^2(X, Y) = (1 - W_2(X, Y)) \cdot H_v^1(X, Y) + W_2(X, Y) \cdot C_2(X, Y, v)$$

$$H_v^3(X, Y) = (1 - W_3(X, Y)) \cdot H_v^2(X, Y) + W_3(X, Y) \cdot C_3(X, Y, v)$$

$$H_v^4(X, Y) = (1 - W_4(X, Y)) \cdot H_v^3(X, Y) + W_4(X, Y) \cdot C_4(X, Y, v)$$

$$H_v^5(X, Y) = (1 - W_5(X, Y)) \cdot H_v^4(X, Y) + W_5(X, Y) \cdot C_5(X, Y, v)$$

$$H_v^6(X, Y) = (1 - W_6(X, Y)) \cdot H_v^5(X, Y) + W_6(X, Y) \cdot C_6(X, Y, v)$$

$$H_v^7(X, Y) = (1 - W_7(X, Y)) \cdot H_v^6(X, Y) + W_7(X, Y) \cdot C_7(X, Y, v)$$

$$H_v^8(X, Y) = (1 - W_8(X, Y)) \cdot H_v^7(X, Y) + W_8(X, Y) \cdot C_8(X, Y, v)$$

$$H_v^9(X, Y) = (1 - W_9(X, Y)) \cdot H_v^8(X, Y) + W_9(X, Y) \cdot C_9(X, Y, v)$$

$$H_v^{10}(X, Y) = (1 - W_{10}(X, Y)) \cdot H_v^9(X, Y) + W_{10}(X, Y) \cdot C_{10}(X, Y, v)$$

$$H_v^{11}(X, Y) = (1 - W_{11}(X, Y)) \cdot H_v^{10}(X, Y) + W_{11}(X, Y) \cdot C_{11}(X, Y, v)$$

$$\begin{aligned}
H_v^{12}(X, Y) &= (1 - W_{12}(X, Y)) \cdot H_v^{11}(X, Y) + W_{12}(X, Y) \cdot C_{12}(X, Y, v) \\
H_v^{13}(X, Y) &= (1 - W_{13}(X, Y)) \cdot H_v^{12}(X, Y) + W_{13}(X, Y) \cdot C_{13}(X, Y, v) \\
H_v^{14}(X, Y) &= (1 - W_{14}(X, Y)) \cdot H_v^{13}(X, Y) + W_{14}(X, Y) \cdot C_{14}(X, Y, v) \\
H_v^{15}(X, Y) &= (1 - W_{15}(X, Y)) \cdot H_v^{14}(X, Y) + W_{15}(X, Y) \cdot C_{15}(X, Y, v) \\
H_v(X, Y) &= H_v^{15}(X, Y)
\end{aligned}$$

Final RGB Channel Evaluation

$$\text{Color}(x, y) = (F(H_0(X, Y)), F(H_1(X, Y)), F(H_2(X, Y)))$$

where $F(x) = \left[255 e^{-e^{-1000x}} |x| e^{-e^{1000(x-1)}} \right]$ performs exact differentiable color clamping to $[0, 255]$.